



Inspiring Excellence

Assignment-02

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Course: CSE422

Section: 18

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Ans. to the ques. no-01

	Laptop (L)		Mobile (M)	
	Student (S)	Teacher (T)	Student (S)	Teacher (T)
High (H)	0.12	0.15	0.2	0.18
Low (Lo)	0.08	0.05	0.07	0.15

a

$$P(T) = 0.15 + 0.05 + 0.18 + 0.15$$
$$= 0.53$$

b

$$P(L_o | M) = \frac{P(L_o \cap M)}{P(M)}$$

$$= \frac{0.07 + 0.15}{0.2 + 0.18 + 0.07 + 0.15}$$
$$= 0.3667$$

Let, User = S ; C Performance = L₀

$$P(S \wedge L_0 | M) = P(S | M) \cdot P(L_0 | M)$$

$$\therefore \text{L.H.S} = P(S \wedge L_0 | M)$$

$$= \frac{P(S \wedge L_0 \wedge M)}{P(M)}$$

$$= \frac{0.07}{0.2 + 0.18 + 0.07 + 0.15}$$

$$= 0.1167$$

$$\therefore \text{R.H.S} = P(S | M) \cdot P(L_0 | M)$$

$$= \frac{P(S \wedge M) \cdot P(L_0 \wedge M)}{\{P(M)\}^2}$$

$$= \frac{(0.2 + 0.07)(0.07 + 0.15)}{(0.2 + 0.18 + 0.07 + 0.15)^2}$$

$$= 0.165$$

As, L.H.S $\neq$ R.H.S

Not conditionally independent.

Ans. to the ques. no-02

a

$$P(\text{Yes} | \text{Evening} \wedge \text{Lakeride} \wedge \text{Barking})$$

$$= \frac{P(\text{Evening} \wedge \text{Lakeride} \wedge \text{Barking} | \text{Yes}) P(\text{Yes})}{P(\text{Evening} \wedge \text{Lakeride} \wedge \text{Barking})}$$

$$= P(\text{Evening} | \text{Yes}) P(\text{Lakeride} | \text{Yes}) P(\text{Barking} | \text{Yes}) P(\text{Yes})$$

$$= \frac{1}{4} \times \frac{2}{4} \times \frac{4}{4} \times \frac{4}{8}$$

$$= 0.0625$$

$$P(\text{No} | \text{Evening} \wedge \text{Lakeride} \wedge \text{Barking})$$

$$= \frac{P(\text{Evening} \wedge \text{Lakeride} \wedge \text{Barking} | \text{No}) P(\text{No})}{P(\text{Evening} \wedge \text{Lakeride} \wedge \text{Barking})}$$

$$= P(\text{Evening} | \text{No}) P(\text{Lakeride} | \text{No}) P(\text{Barking} | \text{No}) P(\text{No})$$

$$= \frac{1}{4} \times \frac{0}{4} \times \frac{1}{4} \times \frac{4}{8}$$

$$= 0.$$

$$\text{Ans. } 0.0625 > 0$$

$\therefore$ Yes, the animal is predicted to belong to a rare species.

Ans. to the ques. no-03

$$P(y=0 | x_1=1 \wedge x_2=1) = \frac{P(x_1=1 \wedge x_2=1 | y=0) P(y=0)}{P(x_1=1 \wedge x_2=1)}$$

$$= P(x_1=1 | y=0) P(x_2=1 | y=0) P(y=0)$$

$$= \frac{1}{3} \times \frac{2}{3} \times \frac{3}{6}$$

$$= \frac{1}{9}$$

$$P(y=1 | x_1=1 \wedge x_2=1) = \frac{P(x_1=1 \wedge x_2=1 | y=1) P(y=1)}{P(x_1=1 \wedge x_2=1)}$$

$$= P(x_1=1 | y=1) P(x_2=1 | y=1) P(y=1)$$

$$= \frac{1}{2} \times \frac{1}{2} \times \frac{2}{6}$$

$$= \frac{1}{12}$$

$$P(y=2 | x_1=1 \wedge x_2=1) = \frac{P(x_1=1 \wedge x_2=1 | y=2) P(y=2)}{P(x_1=1 \wedge x_2=1)}$$

$$= P(x_1=1 | y=2) P(x_2=1 | y=2) P(y=2)$$

$$= \frac{1}{1} \times \frac{1}{1} \times \frac{1}{6}$$

$$= \frac{1}{6}$$

Ans, $\frac{1}{6} > \frac{1}{9} > \frac{1}{12} \therefore$ The most likely outcome is $y=2$

6

Now, if we add the feature x_3 which belongs to R , we can use Gaussian Naive Bayes.

For each y , we calculate

μ_y = mean of x_3 for that class

σ_y^2 = Variance of x_3 for that class

Then we calculate the Gaussian probability density:

$$P(x_3|y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} e^{-\frac{(x_3 - \mu_y)^2}{2\sigma_y^2}}$$

The naive bayes formula becomes:

$$\begin{aligned} P(y|x_1, x_2, x_3) &= \frac{P(x_1, x_2, x_3|y) P(y)}{P(x_1, x_2, x_3)} \\ &= P(x_1|y) P(x_2|y) P(x_3|y) P(y) \end{aligned}$$

Ans. to the ques. no - 04

a

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_{\text{actual}(i)} - y_{\text{pred}(i)})^2$$

Now, as there are two features equation of

y_{pred} looks like, $y = ax_1 + bx_2 + c$

$$\begin{aligned} \therefore MSE &= \frac{1}{4} \left\{ (5 - (ax_1 + bx_2 + c))^2 + (4 - (ax_1 + bx_2 + c))^2 + \right. \\ &\quad \left. (1 - (ax_1 + bx_2 + c))^2 + (9 - (ax_1 + bx_2 + c))^2 \right\} \\ &= \frac{1}{4} \left\{ (5 - (a + 3b + c))^2 + (4 - (2a + 2b + c))^2 + \right. \\ &\quad \left. (1 - (0 + 4b + c))^2 + (9 - (3a + b + c))^2 \right\} \end{aligned}$$

Let, $a = 0.2$; $b = 0.3$; $c = 0.4$

$$\begin{aligned} \therefore MSE &= \frac{1}{4} \{ 12.25 + 6.76 + 0.36 + 59.29 \} \\ &= 19.665 \end{aligned}$$

b

1st iteration:

$$\begin{aligned}\frac{d}{da}(\text{MSE}) &= \frac{1}{4} \left[\left\{ 2(5 - (a + 3b + c)) \times (0 - 1) \right\} \right. \\ &\quad + \left\{ 2(4 - (2a + 2b + c)) \times (0 - 2) \right\} \\ &\quad + \left\{ 2(1 - (4b + c)) \times (0 - 0) \right\} \\ &\quad \left. + \left\{ 2(9 - (3a + b + c)) \times (0 - 3) \right\} \right] \\ &= \frac{1}{4} \left[-2(5 - (a + 3b + c)) - 4(4 - (2a + 2b + c)) \right. \\ &\quad \left. - 6(9 - (3a + b + c)) \right] \dots \dots (i)\end{aligned}$$

$$\text{Let, } a = 0.2; b = 0.3; c = 0.4; \alpha = 0.02$$

$$\frac{d}{da}(\text{MSE}) = -15.9 \quad [\text{From (i)}]$$

$$\begin{aligned}\text{Step size} &= -15.9 \times 0.02 \\ &= -0.318\end{aligned}$$

$$\begin{aligned}a_{\text{new}} &= a_{\text{old}} - \text{stepsize} \\ &= 0.2 + 0.318 \\ &= 0.518\end{aligned}$$

$$\begin{aligned}
 \frac{d}{db}(MSE) &= \frac{1}{4} \left[\{2(5-(a+3b+c)) \times (0-3)\} + \right. \\
 &\quad \{2(4-(2a+2b+c)) \times (0-2)\} + \\
 &\quad \{2(1-(4b+c)) \times (0-4)\} + \\
 &\quad \left. \{2(9-(3a+b+c)) \times (0-1)\} \right] \\
 &= \frac{1}{4} \left[-6(5-(a+3b+c)) - 4(4-(2a+2b+c)) \right. \\
 &\quad \left. - 8(1-(4b+c)) - 2(9-(3a+b+c)) \right] \dots\dots (ii)
 \end{aligned}$$

Let, $a = 0.2$, $b = 0.3$; $c = 0.4$; $\alpha = 0.02$

$$\frac{d}{db}(MSE) = -10.5 \text{ [From. (ii)]}$$

$$\begin{aligned}
 \text{Step size} &= -10.5 \times 0.02 \\
 &= -0.21
 \end{aligned}$$

$$\begin{aligned}
 b_{\text{new}} &= b_{\text{old}} - \text{step size} \\
 &= 0.3 + 0.21 \\
 &= 0.51
 \end{aligned}$$

$$\frac{d}{dc} (MSE) = \frac{1}{4} \left[-2(5 - (a + 3b + c)) - 2(4 - (2a + 2b + c)) \right. \\ \left. - 2(1 - (4b + c)) - 2(9 - (3a + b + c)) \right] \\ \dots\dots\dots(iii)$$

$$\text{Let, } a = 0.2; b = 0.3, c = 0.4 ; \alpha = 0.02$$

$$\frac{d}{dc} (MSE) = -6.6 \quad [\text{From iii}]$$

$$\text{Step size} = -6.6 \times 0.02 \\ = -0.132$$

$$C_{\text{new}} = C_{\text{old}} - \text{step size} \\ = 0.4 + 0.132 \\ = 0.532$$

2nd iteration:

$$a = 0.518 ; b = 0.51 ; c = 0.532 ; \alpha = 0.02$$

$$\frac{d}{da}(\text{MSE}) = -12.228 \quad [\text{From (i)}]$$

$$\begin{aligned}\text{Stepsize} &= -12.228 \times 0.02 \\ &= -0.2446\end{aligned}$$

$$\begin{aligned}\therefore a_{\text{new}} &= a_{\text{old}} - \text{Stepsize} \\ &= 0.7626\end{aligned}$$

$$\frac{d}{db}(\text{MSE}) = -5.1 \quad [\text{From (ii)}]$$

$$\begin{aligned}\text{Stepsize} &= -5.1 \times 0.02 \\ &= -0.102\end{aligned}$$

$$\begin{aligned}b_{\text{new}} &= b_{\text{old}} - \text{Stepsize} \\ &= 0.612\end{aligned}$$

$$\frac{d}{dc}(\text{MSE}) = -4.332 \quad [\text{From (iii)}]$$

$$\text{Stepsize} = -0.0866$$

$$\begin{aligned}c_{\text{new}} &= c_{\text{old}} - \text{Stepsize} \\ &= 0.6186\end{aligned}$$

C

$$y = ax_1 + bx_2 + C$$

$$= 0.7626x_1 + 0.612x_2 + 0.6186$$

$$x_1 = 2.5 \text{ hours} ; x_2 = 2.5 \text{ hours}$$

$$\therefore y = 4.0551$$

Ans. to the ques. no-05

$$L = BCE = -\frac{1}{n} \sum_{i=1}^n [y_i \log(a_i) + (1-y_i) \log(1-a_i)];$$

$\left[\begin{array}{l} y = \text{actual} \\ a = \text{predicted} \end{array} \right]$

$$a = \frac{1}{1 + e^{-z}} \quad ; \quad z = w_1 x_1 + w_2 x_2 + c$$

$$\text{Let, } w_1 = 0.2; w_2 = 0.3; c = 0.4$$

For sample-1:

$$\begin{aligned} z &= 0.2 \times 6.5 + 0.3 \times 3 + 0.4 \\ &= 2.6 \end{aligned}$$

$$\therefore a_1 = \frac{1}{1 + e^{-2.6}} = 0.9309$$

For sample-2:

$$\begin{aligned} z &= 0.2 \times 6.2 + 0.3 \times 0 + 0.4 \\ &= 1.64 \end{aligned}$$

$$\therefore a_2 = 0.8375$$

For sample-3:

$$\begin{aligned} z &= 0.2 \times 5 + 0.3 \times 4 + 0.4 \\ &= 2.6 \end{aligned}$$

$$a_3 = 0.93086$$

$$\begin{aligned}
 BCE &= -\frac{1}{3} [0 \log(0.9309) + (1-0) \log(1-0.9309) \\
 &\quad + 1 \log(0.8375) + (1-1) \log(1-0.8375) + \\
 &\quad + 0 \log(0.93086) + (1-0) \log(1-0.93086)] \\
 &= -\frac{1}{3} (-2.67 - 0.177 - 2.67) \\
 &= 1.839
 \end{aligned}$$

1st iteration:

b

Sample	Cholesterol level (x_1)	Exercise Hour (x_2)	Has Disease (y)	$(a-y)x_1$	$(a-y)x_2$	$(a-y)$
1	6.5	3	0	6.0508	2.7927	0.9309
2	6.2	0	1	-1.0075	0	-0.1625
3	5.0	4	0	4.6543	3.7234	0.93086

$$\begin{aligned}
 \therefore \frac{d}{d\omega_1}(L) &= \frac{1}{3} \times (6.0508 - 1.0075 + 4.6543) \\
 &= 3.2325
 \end{aligned}$$

$$\begin{aligned}
 \therefore \frac{d}{d\omega_2}(L) &= \frac{1}{3} (2.7927 + 0 + 3.7234) \\
 &= 2.172
 \end{aligned}$$

$$\begin{aligned}
 \therefore \frac{d}{d\omega_3}(L) &= \frac{1}{3} (0.9309 - 0.1625 + 0.93086) \\
 &= 0.5664
 \end{aligned}$$

$$\omega_{1, \text{new}} = 0.2 - (3.2325 \times 0.02)$$

$$= 0.13535$$

$$\omega_{2, \text{new}} = 0.3 - (2.172 \times 0.02)$$

$$= 0.25656$$

$$C_{\text{new}} = 0.4 - (0.5664 \times 0.02)$$

$$= 0.38867$$

2nd iteration:

$$\omega_1 = 0.13535 ; \omega_2 = 0.25656 ; C = 0.38867$$

For sample-1:

$$z = 0.13535 \times 6.5 + 0.25656 \times 3 + 0.38867$$

$$= 2.038$$

$$\therefore a = \frac{1}{1 + e^{-2.038}} = 0.8897$$

For sample-2:

$$z = 0.13535 \times 6.2 + 0.25656 \times 0 + 0.38867$$

$$= 1.228$$

$$\therefore a = 0.7735$$

For sample - 3:

$$\bar{z} = 0.13535 \times 5 + 0.25656 \times 4 + 0.38867$$

$$= 2.0917$$

$$a = 0.890$$

Sample	Cholesterol level (x_1)	Exercise hours (x_2)	Has disease (y)	$(a-y)x_1$	$(a-y)x_2$	$(a-y)$
1	6.5	3	0	5.75	2.6541	0.8847
2	6.2	0	1	-1.404	0	-0.2265
3	5.0	4	0	4.45	3.56	0.890

$$\frac{d}{d\omega_1}(L) = \frac{1}{3}(5.75 - 1.404 + 4.45)$$

$$= 2.932$$

$$\frac{d}{d\omega_2}(L) = \frac{1}{3}(2.6541 + 0 + 3.56)$$

$$= 2.0714$$

$$\frac{d}{dc}(L) = \frac{1}{3}(0.8847 - 0.2265 + 0.890)$$

$$= 0.5161$$

$$\omega_{1\text{ new}} = 0.13535 - (2.932 \times 0.02)$$

$$= 0.07671$$

$$\omega_{2\text{ new}} = 0.25656 - (2.0714 \times 0.02)$$

$$= 0.21513$$

$$c_{\text{new}} = 0.38867 - (0.5161 \times 0.02)$$

$$= 0.3783$$

C

$$\begin{aligned} \hat{z} &= 0.07671x_1 + 0.21513x_2 + 0.3783 \\ &= 0.07671 \times 4 + 0.21513 \times 1 + 0.3783 \\ &= 0.9003 \end{aligned}$$

$$\begin{aligned} a &= \frac{1}{1 + e^{-0.9003}} \\ &= 0.7110 \end{aligned}$$

$$\text{Ans } a \geq 0.5$$

$$\text{or } 0.7110 > 0.5$$

$\therefore$ The person is predicted to have disease.

Ans. to the ques. no - 06

a

ReLU:

$$f(z) = \max(0, z) = \begin{cases} z, & \text{if } z \geq 0 \\ 0, & z < 0 \end{cases}$$

Let, the top hidden neuron be h_1 .

$$\begin{aligned} z_1 &= w_1 x_1 + w_2 x_2 \\ &= 4 \times 5 + (-1) \times 1 \\ &= 20 - 1 \\ &= 19 \end{aligned}$$

$$h_1 = \max(0, 19) = 19$$

Let, the bottom hidden layer be h_2 .

$$\begin{aligned} z_2 &= w_1 x_1 + w_2 x_2 \\ &= 3 \times 5 + 2 \times 1 \\ &= 15 + 2 \\ &= 17 \end{aligned}$$

$$h_2 = \max(0, 17) = 17$$

Input of the output neuron:

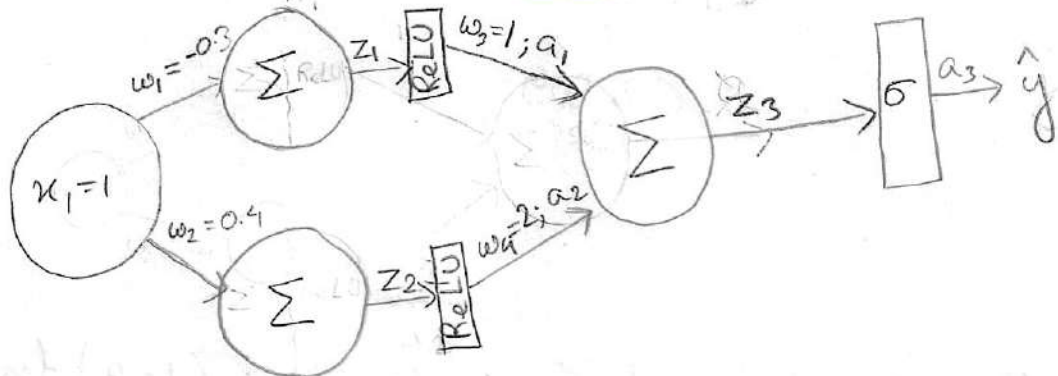
$$\begin{aligned} z_y &= h_1 \omega_1 + h_2 \omega_2 \\ &= 19 \times 3 + 17 \times 2 \\ &= 91 \end{aligned}$$

$$\begin{aligned} \therefore y &= \frac{1}{1 + e^{-91}} \\ &= 1 \end{aligned}$$

$$\text{Ans: } Y = 1$$

Ans. to the ques. no-07

a



b

$$z_1 = w_1 x_1$$

$$= -0.3 \times 1$$

$$= -0.3$$

$$a_1 = \max(0, -0.3)$$

$$= 0$$

$$z_2 = w_2 x_1$$

$$= 0.4 \times 1$$

$$= 0.4$$

$$a_2 = \max(0, 0.4)$$

$$= 0.4$$

$$\therefore \text{Input of output neuron} = z_3 = a_1 w_3 + a_2 w_4$$

$$= 1 \times 0 + 0.4 \times 2$$

$$z_3 = 0.8$$

$$\therefore Q_3 = \frac{1}{1 + e^{-0.8}} = \frac{1}{1 + e^{-0.83}} = \frac{1}{1 + e^{-a_1 \omega_3 + a_2 \omega_4}}$$

$$= 0.6899$$

C

$$BCE = L = -\frac{1}{n} \{ y \ln(a) + (1-y) \ln(1-a) \}$$

$$= -\frac{1}{1} \{ 1 \ln(0.6899) + (1-0.6899) \ln(1-0.6899) \}$$

$$= -\{ \ln(0.6899) \}$$

$$= 0.3712$$

$$L = -\frac{1}{n} \sum_{i=1}^n \{y_i \ln(a_i) + (1-y_i) \ln(1-a_i)\}$$

$$= -\frac{1}{n} \sum_{i=1}^n \{y \ln(a_3) + (1-y) \ln(1-a_3)\}$$

$$\frac{d}{d\omega_4}(L) = -\frac{d}{da_3}(L) * \frac{d(a_3)}{d\beta_3} * \frac{d(\beta_3)}{d\omega_4}$$

$$= -\left(\frac{y}{a_3} - \frac{1-y}{1-a_3}\right) \times \{-1(1+e^{-\beta_3})^2(-1 \times e^{-\beta_3})\}$$

$$\times a_2$$

$$= -\frac{y(1-a_3) - a_3(1-y)}{a_3(1-a_3)} \times \left(a_3^2 \times \left\{\frac{1}{a_3} - 1\right\}\right) \times a_2$$

$$= -\frac{y(1-a_3) - a_3(1-y)}{a_3(1-a_3)} \times \left(a_3^2 \times \frac{1-a_3}{a_3}\right) \times a_2$$

$$= -(y(1-a_3) - a_3(1-y)) a_2$$

$$= -(y - a/y - a_3 + a/y) a_2$$

$$= (a_3 - y) a_2$$

$$\frac{d}{d\omega_3}(L) = \frac{d}{da_3}(L) \times \frac{d}{dz_1}(a_3) \times \frac{d}{d\omega_3}(z_3)$$

$$= (a_3 - y) a_1$$

$$\frac{d}{d\omega_2}(L) = \frac{d}{da_3}(L) \times \frac{d}{dz_3}(a_3) \times \frac{d}{da_2}(z_3) \times \frac{da_2}{dz_2} \times \frac{dz_2}{d\omega_2}$$

$$= (a_3 - y) \omega_4 [\text{ReLU}'(z_2)] \times n_1$$

$$\frac{d}{d\omega_1}(L) = \frac{d}{da_3}(L) \times \frac{d}{dz_3}(a_3) \times \frac{d}{da_1}(z_3) \times \frac{da_1}{dz_1}$$

$$\times \frac{d}{dz_1}(a_1) \times \frac{d}{d\omega_1}(z_1)$$

$$= (a_3 - y) \omega_3 [\text{ReLU}'(z_1)] n_1$$

We know, for ReLU:

$$f(z) = \max(0, z) = \begin{cases} z; & \text{if } z \geq 0 \\ 0; & \text{if } z < 0 \end{cases}$$

$$\text{Derivative: } f'(z) = \begin{cases} 1; & \text{if } z > 0 \\ 0; & \text{if } z \leq 0 \end{cases}$$

$$\omega_{1, \text{new}} = -0.3 - (0.1 \times 0)$$

$$= -0.3$$

$$\omega_{2, \text{new}} = 0.4 - \{0.1 \times (0.6899 - 1) \times 2 \times 1\}$$

$$= 0.46202$$

$$\omega_{3, \text{new}} = 1 - \{0.1 \times (0.6899 - 1) \times 0\}$$

$$= 1$$

$$\omega_{4, \text{new}} = 2 - \{0.1 \times (0.6899 - 1) \times 0.4\}$$

$$= 2.0124$$

Ans. to the ques. no-08

a

$$L = - [y \log(a_3) + (1-y) \log(1-a_3)]$$

$$\therefore L = -\ln(a_3)$$

In general,

$$L = -\frac{1}{n} \sum_{i=1}^n [y_i \log(a_i) + (1-y_i) \log(1-a_i)]$$

b

$$\begin{aligned} \bar{z}_1 &= x_1 \omega_1 + x_2 \omega_2 + \beta_1 \\ &= 1 \times 0.1 + 2 \times 0.5 + 0.5 \\ &= 1.6 \end{aligned}$$

$$a_1 = \frac{1}{1 + e^{-1.6}} = 0.832$$

$$\begin{aligned} \bar{z}_2 &= a_1 \omega_3 + \beta_2 \\ &= 0.832 \times 0.5 + 0.5 \\ &= 0.916 \end{aligned}$$

$$a_2 = \frac{1}{1 + e^{-0.916}} = 0.71422$$

$$\bar{z}_3 = a_2 \omega_4 + \beta_3$$

$$= 0.71422 \times 0.2 + 0.5$$

$$= 0.6428$$

$$\therefore a_3 = \frac{1}{1 + e^{-0.6428}} = 0.6554$$

C

$$\frac{d}{d\omega_4}(L) = \frac{d}{da_3}(L) * \frac{d}{da_3}(\bar{z}_3) \times \frac{d}{d\bar{z}_3}(\omega_4)$$

$$\frac{d}{d\omega_4}(L) = \frac{d}{da_3}(L) \times \frac{d}{d\bar{z}_3}(a_3) \times \frac{d}{d\omega_4}(\bar{z}_3)$$

$$= -\left(\frac{y}{a_3} - \frac{1-y}{1-a_3}\right) \times \left\{ -1(1 + e^{-\bar{z}_3})^2 \times e^{-\bar{z}_3} \times (-1) \right\} \times a_2$$

$$= -\frac{y(1-a_3) - (1-y)a_3}{a_3(1-a_3)} \times \left(a_3^2 \times \left\{ \frac{1}{a_3} - 1 \right\} \right) a_2$$

$$= -\frac{y(1-a_3) - (1-y)a_3}{a_3(1-a_3)} \times \left\{ a_3^2 \times \frac{1-a_3}{a_3} \right\} a_2$$

$$= -\{ y(1-a_3) - (1-y)a_3 \} a_2$$

$$= -\{ y - a_3 y - a_3 + a_3 y \} a_2$$

$$= (a_3 - y) a_2$$

$$\frac{d(L)}{d\omega_4} = (0.6554 - 1) \times 0.71422$$

$$= -0.2461$$

$$\frac{d}{d\omega_1}(L) = \frac{d}{da_3}(L) \times \frac{d}{dz_3}(a_3) \times \frac{d}{da_2}(z_3) \times \frac{d}{dz_2}(a_2)$$

$$\times \frac{d}{da_1}(z_2) \times \frac{d}{dz_1}(a_1) \times \frac{d}{dz_0}(z_1)$$

$$= (a_3 - y) \omega_4 \left\{ (1 + e^{-z_2})^{-2} e^{-z_2} \right\} \omega_3 \times$$

$$\left\{ (1 + e^{-z_1})^{-2} e^{-z_1} \right\} \times u_1$$

$$= (a_3 - y) \omega_4 \left\{ a_2 (1 - a_2) \right\} \omega_3 \left\{ a_1 (1 - a_1) \right\} u_1$$

$$= -0.000983$$

Assignment 3
CSE422: Artificial Intelligence
Total Marks – 20

Q1. Consider the joint distribution table of three random variables.

Device	User	Performance	Probability
Laptop	Student	High	0.12
Laptop	Student	Low	0.08
Laptop	Teacher	High	0.15
Laptop	Teacher	Low	0.05
Mobile	Student	High	0.2
Mobile	Student	Low	0.07
Mobile	Teacher	High	0.18
Mobile	Teacher	Low	0.15

Answer the following questions –

- Calculate the marginal probability of a user being a teacher.
- If you are given a mobile device, calculate the probability of it having low performance.
- Are User and Performance conditionally independent of each other, given Mobile?

Q2. A national park uses sensors and ranger observations to classify whether an animal sighting belongs to a rare species.

ID	Day Period	Location	Sound	Rare
A01	Morning	Lakeside	Yes	Yes
A02	Night	Deep forest	No	No
A03	Evening	Deep forest	Yes	Yes
A04	Morning	Grassland	No	No
A05	Night	Lakeside	Yes	Yes
A06	Evening	Grassland	No	No
A07	Morning	Deep forest	Yes	Yes
A08	Night	Grassland	Yes	No

Answer the following questions –

- Predict whether the animal you saw last evening, at lakeside, and heard barking belongs to a rare species.

Q3. In the following dataset, x_1 and x_2 are features, and y belongs to a multinomial distribution:

x_1	x_2	y
0	1	0
1	0	0
1	1	1
0	1	0
0	0	1
1	1	2

Answer the following questions -

- Using Naive Bayes, calculate the most likely outcome of y given $x_1 = 1$ and $x_2 = 1$.
- Consider that a new feature x_3 with continuous values, that is, $x_3 \in \mathbb{R}$, is to be added to the dataset. How can the naive Bayes classifier be applied with x_3 ?

Q4. Consider the following regression dataset:

Sample	Studying Hours (x_1)	Hours Sleeping (x_2)	Test score (y)
1	1	3	5
2	2	2	4
3	0	4	1
4	3	1	9

You decide to use linear regression to solve this problem. For the weights, choose any value between $[0.1, 0.9]$. The learning rate must be between $[0.01, 0.1]$.

Answer the following questions -

- Calculate the Mean Squared Error (MSE) for this model on the dataset.
- Perform two iterations to update the parameters.
- Using the updated values, find out the predicted test score for a student who studied 2.5 hours and slept 2.5 hours.

Q5. Consider the following classification dataset:

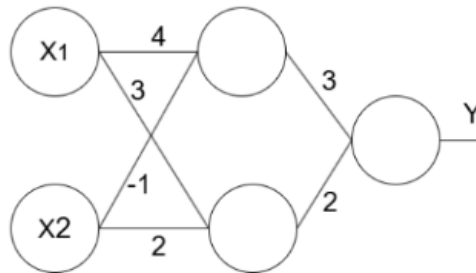
Sample	Cholesterol level (x_1)	Exercise hour (x_2)	Has Disease? (y)
1	6.5	3	0
2	6.2	0	1
3	5.0	4	0

You decide to use logistic regression to solve this problem. For the weights, choose any value between [0.1, 0.9]. The learning rate must be between [0.01, 0.1].

Answer the following questions -

- Calculate the error using Binary Cross-Entropy (BCE) for this model on the dataset.
- Perform two iterations to update the parameters.
- Using the updated values, predict whether a person with a cholesterol level of 4.0 and 1 hour of exercise will have the disease.

Q6. Consider the following feed-forward neural network. Here, the circles represent a Perceptron (neuron), which is a weighted sum followed by an activation function. The hidden-layer neurons use ReLU activation, whereas the output-layer neurons use sigmoid activation.



Answer the following questions -

- If $x_1 = 5$ and $x_2 = 1$, apply forward propagation to calculate the value of y .

Q7. Suppose we are training a Feedforward Neural Network with a single feature. Network structure is as follows:

Input feature, $x_1 = 1$

Hidden layer: 2 Neurons, weights are - 0.3 and 0.4, ReLU activation

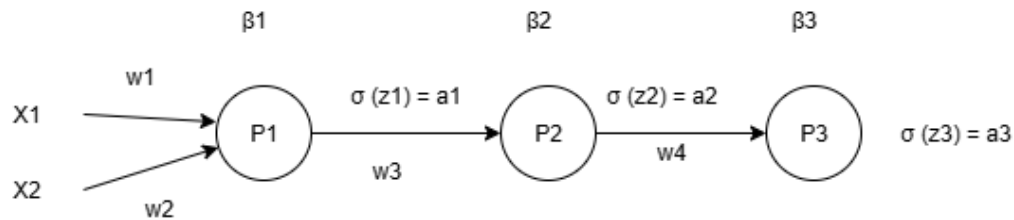
Output layer: 1 neuron, weights are 1 and 2, Sigmoid activation

Loss function: Binary Cross-Entropy

Answer the following questions -

- Draw the neural network.
- Run the forward propagation and find the output.
- If given output $y = 1$, calculate the error using the loss function.
- Update all weights in the neural network using back propagation. You must show the derivatives. Take learning rate = 0.1.

Q8.



Suppose you are given the architecture above as an MLP for an inferential task. Your test set appears to be $x_1 = 1$, $x_2 = 2$, and $y = 1$ (true label). Furthermore, the network assumes the weights and biases are initialized as $w_1 = 0.1$, $w_2 = 0.5$, $w_3 = 0.5$, $w_4 = 0.2$, $\beta_1 = 0.5$, $\beta_2 = 0.5$, and $\beta_3 = 0.5$, respectively, as part of its initial hypothesis. Moreover, P1, P2, and P3 are the three Perceptrons whose non-linear outputs are a_1 , a_2 , and y' , respectively. The Loss function is Binary Cross-Entropy.

Answer the following questions -

- Write the equation of the loss function.
- Perform one pass of forward propagation and determine the predicted output for the given test set.
- For one pass of backward propagation, first derive the expression for $\partial L / \partial w_4$. Then use the chain rule to derive the expression for $\partial L / \partial w_1$. Finally, calculate the quantitative values.

Q9. A bank seeks to predict whether a customer will default on a loan (Yes/No) based on their credit score (Low, Medium, High) and previous loan history (Good, Bad). Altogether, the dataset has records of 100 people, with the following distribution:

- 45 people with Low credit scores: 15 defaults, 25 do not default.
- 30 people with a medium credit score: 10 defaults, 20 do not default.
- 20 people with High credit scores: 5 defaults, 15 do not default.
- 50 people with good loan history: 5 defaults, 45 do not default.
- 50 people with Bad loan history: 25 defaults, 25 do not default.

Answer the following questions -

- Calculate the entropy of the dataset.
- What is the information gained if we split the dataset based on credit score?
- What is the information gain if we split the dataset based on load history?

Q10. Decision tree construction. For the following dataset, construct the full decision tree using the ID3 algorithm shown in class.

Attributes:

- Age Group: Young (Y), Middle-aged (M), Senior (S)
- Employment Status: Employed (E), Unemployed (U)
- Interest in Community Events: Low (L), Medium (M), High (H)

Target Variable: Participation (Yes, No)

Age Group	Employment Status	Interest in Community Events	Participation
Young	Employed	High	Yes
Young	Employed	Medium	Yes
Young	Unemployed	Low	No
Young	Unemployed	High	Yes
Middle-aged	Employed	Medium	Yes
Middle-aged	Employed	Low	No
Middle-aged	Unemployed	Medium	No
Senior	Employed	High	Yes
Senior	Unemployed	Low	No
Senior	Unemployed	Medium	No
Senior	Unemployed	Low	No
Young	Employed	Low	No
Middle-aged	Employed	High	Yes
Senior	Unemployed	High	Yes

Hint:

- Before attempting this problem, make sure you understand the situation shown in the slide thoroughly.
- First calculate $\text{Gain}(\text{Participation}, \text{Age Group})$, $\text{Gain}(\text{Participation}, \text{Employment})$, $\text{Gain}(\text{Participation}, \text{Interest})$. Choose the attribute with the highest gain.
- Suppose Employment Status has the highest gain. Choose this as the root of the tree. Since Employment Status has two classes (employed, unemployed), the root will have two children. We now need to choose the following attribute.
- Calculate $\text{Gain}(\text{Age Group}, \text{employed})$ and $\text{Gain}(\text{Interest}, \text{employed})$. Choose the one with the highest gain.
- Similarly, calculate $\text{Gain}(\text{Age Group}, \text{unemployed})$ and $\text{Gain}(\text{Interest}, \text{unemployed})$. Choose the one with the highest gain.]
- Continue in this way until you reach the leaf nodes (yes or no).